

Question 2

(a)

$$f(x, y, z) = e^{x^2+y^2} \sin(y^2 + xz^2) + z$$

$$\frac{\partial f}{\partial x} = 2xe^{x^2+y^2} \sin(y^2 + z^2x) + e^{x^2+y^2} z^2 \cos(y^2 + z^2x)$$

$$\frac{\partial f}{\partial y} = 2ye^{x^2+y^2} \sin(y^2 + z^2x) + e^{x^2+y^2} 2y \cos(y^2 + z^2x)$$

$$\frac{\partial f}{\partial z} = e^{x^2+y^2} 2z \cos(y^2 + z^2x) + 1$$

(b)

$$f(\rho, \phi, \theta) = \rho^2 \cos \theta \sin^2 \phi$$

$$\frac{\partial f}{\partial \rho} = 2\rho \cos \theta \sin^2 \phi$$

$$\frac{\partial f}{\partial \phi} = \rho^2 \cos \theta 2 \cos \phi \sin \phi$$

$$\frac{\partial f}{\partial \theta} = -\rho^2 \sin \theta \sin^2 \phi$$

(c)

$$f(x, y, z) = x \ln(x^2 z^2 + g^2(y^2 z^3))$$

$$\frac{\partial f}{\partial x} = \ln(x^2 z^2 + g^2(y^2 z^3)) + 2xz^2 \frac{x}{x^2 z^2 + g^2(y^2 z^3)}$$

$$\frac{\partial f}{\partial y} = x \frac{4z^3 y g'(y^2 z^3) g(y^2 z^3)}{x^2 z^2 + g^2(y^2 z^3)}$$

$$\frac{\partial f}{\partial z} = x \frac{2zx^2 + (6y^2 z^2 g'(y^2 z^3) g(y^2 z^3))}{x^2 z^2 + g^2(y^2 z^3)}$$

Question 3

$$\lim_{(x,y,z) \rightarrow (0,0,0)} \frac{xyz}{x^2 + y^4 z^4}$$

- Along the curve $x = t, y = t, z = t$

$$\lim_{t \rightarrow 0} \frac{t^3}{t^2 + 2t^4} = 0$$

- Along the curve $x = t, y = \sqrt{t}, z = \sqrt{t}$

$$\lim_{t \rightarrow 0} \frac{t^2}{3t^2} = \frac{1}{3}$$

Since these limits are not equal to each other, we can conclude that f is not continuous at $(0, 0, 0)$

Question 4

$$f(x, y) = \cos(\pi x^3 y^4)$$

(a)

$$f_x = -3\pi x^2 y^4 \sin(\pi x^3 y^4)$$

$$f_y = -4\pi y^3 x^3 \sin(\pi x^3 y^4)$$

(b)

This is simply the curve with the x value fixed at $x = 2$

$$z = \cos(8\pi y^4)$$

(c)

To find the tangent line to the curve at the point $(2, 1, 1)$, we need the slope of the line at this point, which we can use the partial derivative f_y

$$f_y(2, 1) = 0$$

Thus the tangent line is simply

$$\vec{r}(t) = (2, 1, 1) + (0, 1, 0)t$$